

Yichuan Song

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EDUCATION

Bachelor of Engineering **The University of Hong Kong** **Sep 2021 - Jun 2026(expected)**
Computer Science (GPA: 3.55/4.30)

RESEARCH INTERESTS

Energy-Based Models, Long-Trajectory Reasoning & Planning, Mechanistical Interpretability

RECENT RESEARCH WORKS

Debra: A Discrete Method for Energy-Based Reasoning Mar 2025 – present

First Author; Supervised by Yilun Du, Harvard University; Prepare for ICML2026([paper](#), [code](#))

- Proposed a novel annealing EBM training paradigm featuring scheduled mask propagation and contrastively augmented cross-entropy loss with pseudolikelihood.
- Implemented parallel training and sampling of the paradigm and conducted evaluations on discrete reasoning tasks, such as Countdown and Sudoku.

Introspective Reasoning: Does LLM Say What It Thinks? Sep 2024 – Mar 2025

First Author; Supervised by Lingpeng Kong, University of Hong Kong; Draft for ACL2025([paper](#), [code](#))

- Developed an entropy-guided beam-search decoding method for agentic planning task on AgentBoard.
- Developed a probing-intervention LRM decoding pipeline featuring plug-and-play probes and interveners detecting and adjusting the hidden states of the weak reasoning steps during long-CoT reasoning.
- Developed a test-time LRM reasoning pipeline to first detect the attention-agnostic steps and then re-prompt the LRM to internalize the intermediate calculations.

Knowledge Verification on Transformers: Case of Othello May 2024 – Aug 2024

Independent Research Assistant; Supervised by Xujie Si, The University of Toronto; ([slides](#), [code](#))

- Derived a formal upper bound on robustness of the decoder and probes of OthelloGPT via probing and SMT-engined verification and verified that model depth monotonically enlarges the provable bound.
- Implemented parallel retraining of OthelloGPT with BatchNorm replacing LayerNorm.
- Presented two orals of a survey on neural activations and the project.

INTERNSHIP AND EXCHANGES

Research Internship(Globalink, Mitacs) May 2024 - Aug 2024

Bahen Centre for Information Technology, University of Toronto

Summer Session Visitor(HKU SSAP scholarship) Jun 2023 - Aug 2023

University of California, Berkeley *A Grades(CS188 and MATH N53)*

Summer School Visitor(HKU China Vision Programme) Jun 2022 - Aug 2022

Peking University

SERVICE

Student TA for AILT1001: Artificial Intelligence Literacy <i>The University of Hong Kong</i>	2025-26 A.Y.
Student Helper for HKU Archaeology Team <i>Vedi Fortress, Armenian Araxes River Valley</i>	Feb 2023 - May 2023
Member of HKU RoboMaster Team(Mechanical Group) <i>The University of Hong Kong</i>	Sep 2021 - Jan 2023
Student TA for ENGG1340/COMP2113: Computer Programming II <i>The University of Hong Kong</i>	Sep 2022 - Dec 2022

AWARDS

Dean's Honours List	2021-22 A.Y.
2nd Prize in RoboMaster University Championship 2022	Aug 2022
3rd Prize in RoboMaster University League 2022	Aug 2022